THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the roots of $x^2 - 7x + 10 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 69 **(b)** 39 **(c)** 49 **(d)** 29

2. If the roots of $x^2 - 10x + k = 0$ differ by 2, then $k =$ 1

(a) 24 **(b)** 26 **(c)** 23 **(d)** 48

3. For which value of k does $x^2 + kx + 9 = 0$ have equal roots (take $k > 0$)? 1

(a) 12 **(b)** 9 **(c)** 6 **(d)** 3

4. If $x = 1$ is a root of $kx^2 + 3x - 5 = 0$, then $k =$ 1

(a) -8 **(b)** 2 **(c)** 5 **(d)** -2

5. The equation $kx^2 + 6x + 3 = 0$ has real roots when: 1

(a) $k < 3$ only **(b)** $k \geq 3$

(c) $k \leq 3$ and $k \neq 0$ **(d)** $k > 0$

6. If α, β are roots of $x^2 - 5x + 6 = 0$, then $\frac{1}{\alpha} + \frac{1}{\beta} =$ 1

(a) 30 **(b)** $\frac{5}{6}$ **(c)** $\frac{6}{5}$ **(d)** $-\frac{5}{6}$

7. If the roots of $x^2 - 9x + 14 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 81 **(b)** 67 **(c)** 109 **(d)** 53

8. If the roots of $x^2 - 12x + k = 0$ differ by 4, then $k =$ 1

(a) 40 **(b)** 64 **(c)** 128 **(d)** 32

9. **Assertion (A):** The equation $2x^2 - 3x + 5 = 0$ has no real roots. 1

Reason (R): If the discriminant $b^2 - 4ac < 0$, a quadratic equation has no real roots.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The equation $x^2 - 6x + 9 = 0$ has two equal real roots. 1
Reason (R): A quadratic equation always has two distinct real roots.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the values of k for which $kx^2 + 4x + 1 = 0$ has real and equal roots. 2

12. If α, β are the roots of $x^2 - 6x + 5 = 0$, find $\alpha^2 + \beta^2$. 2

13. The sum of a number and its reciprocal is $\frac{5}{2}$. Find the number. 2

14. Find the values of k for which $kx^2 + 6x + 3 = 0$ has real and equal roots. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve for x: $\dfrac{x+1}{x-1} + \dfrac{x-1}{x+1} = \frac{10}{3}$. 3

16. For what values of k does $x^2 + 5x + k = 0$ have real roots? Find the greatest integer value of k . 3

17. The sum of the squares of two consecutive odd positive integers is 74. Find them. 3

OR

Solve for x: $\dfrac{x+2}{x-2} + \dfrac{x-2}{x+2} = \frac{26}{5}$. 3

18. For what values of k does $x^2 + 7x + k = 0$ have real roots? Find the greatest integer value of k . 3

OR

The sum of the squares of two consecutive odd positive integers is 130. Find them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A rectangular park has an area of 180 square metres. Its length is 3 m more than its breadth. Let the breadth be x metres. 5
(i) Form the quadratic equation in x.
(ii) Solve it by factorisation.
(iii) Find the length and breadth of the park.
(iv) A fence of 3 rounds of wire is put around the park. Find the total length of wire and its cost at ₹ 20 per metre.

20. Two taps running together fill a tank in $\frac{24}{5}$ hours. The larger tap alone takes 4 hours less than the smaller one. Find the time each takes alone. 5

OR

A motorboat goes 60 km upstream and 60 km downstream in 8 hours. If the speed of the stream is 4 km/h, find the speed of the boat in still water. 5

END OF QUESTION PAPER — Total: 40 Marks